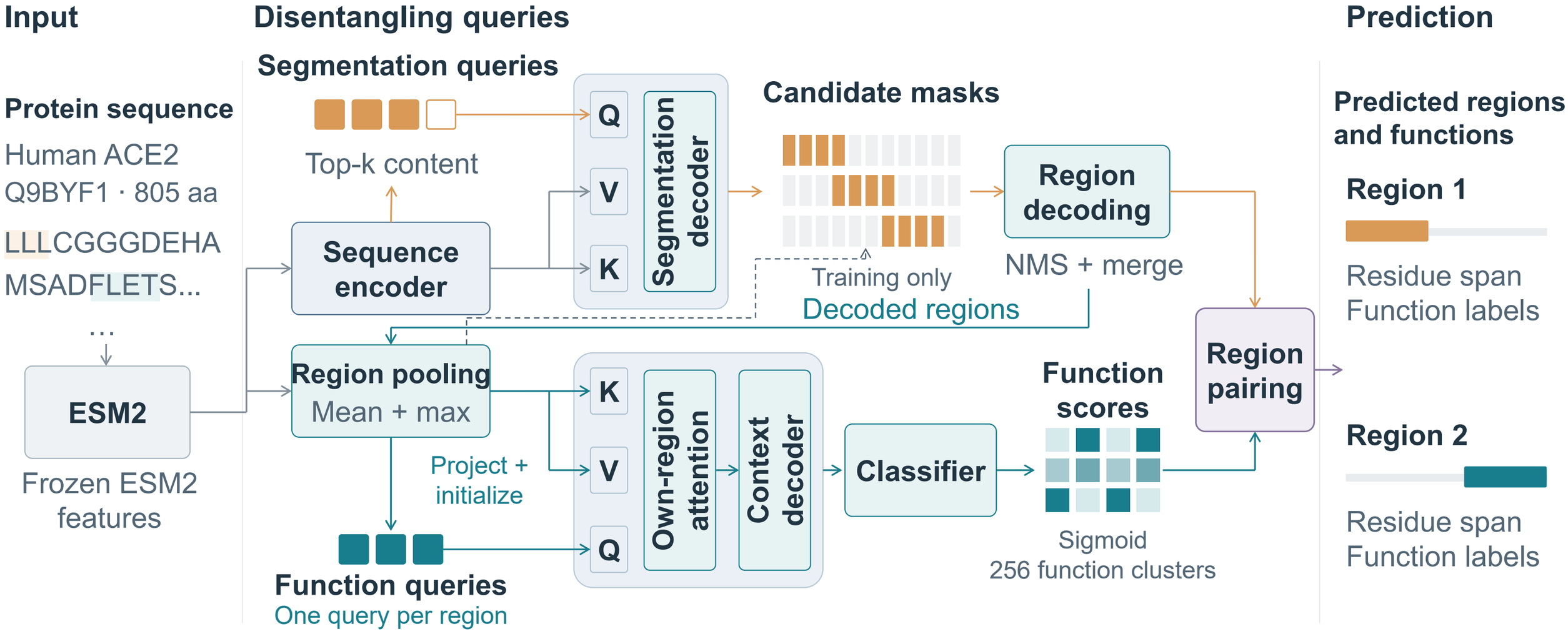

Input
Disentangling queries
Prediction
Protein sequence
Human ACE2
Q9BYF1 · 805 aa
LLLCGGGDEHA
MSADFLETS...
...
ESM2
Frozen ESM2 features
Segmentation queries
Top-k content
Sequence encoder
Q
V
K
Segmentation decoder
Candidate masks
Training only
Region decoding
NMS + merge
Decoded regions
Region pooling
Mean + max
Project + initialize
Function queries
One query per region
K
V
Q
Own-region attention
Context decoder
Classifier
Function scores
Sigmoid
256 function clusters
Region pairing
Predicted regions and functions
Region 1
Residue span
Function labels
Region 2
Residue span
Function labels